Some Basic Statistical Tools

Outline for Today

1. Note on methods of learning
2. Convergence for sequences of random variables
3. Break
4. Concentration inequalities
5. Proof of the weak law of large numbers
6. (If time) Proof techniques

Methods of Learning

(Disclaimer: I am not a trained epistemologist)

Testimony

1. Read or hear a piece of knowledge
2. Decide whether to believe it (e.g. based on authority of speaker)
3. Learn by believing it

Perception (Scientific Method)

1. Formulate a question
2. Formulate a hypothesis
3. Test the hypothesis with an experiment
4. Learn by analyzing the results of the experiment

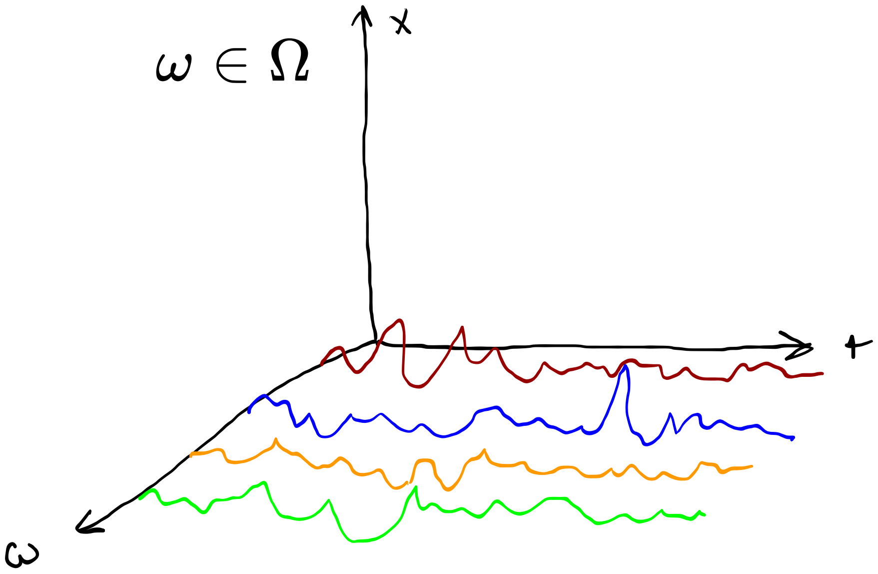
Reason (Mathematical Proof)

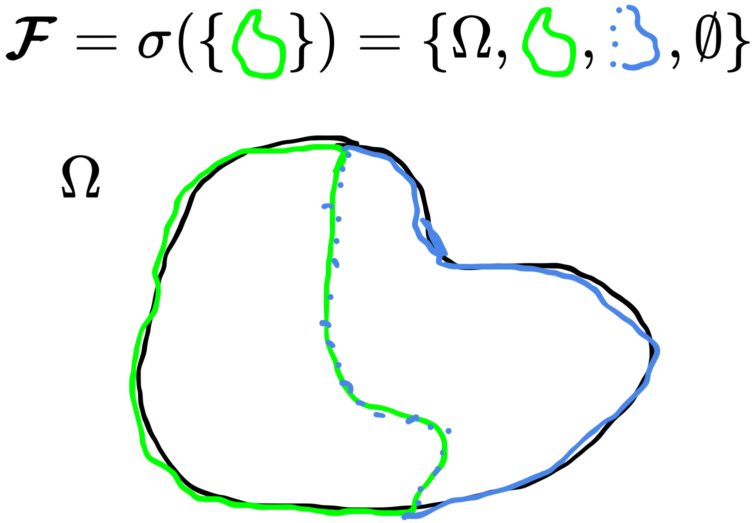
1. Formulate a question
2. Create a conjecture that answers the question
   1. *Define* all concepts
   2. State as a theorem
3. Prove or disprove the conjecture
4. Learn by considering the theorem/false conjecture and the proof

Most RL research fits into the scientific method.

Review

Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E-valued **random variable is a measurable function** $X : \Omega \to E$.





Review: Equality of Random Variables

Review: Convergence

For a (deterministic) sequence $\{x_n\}$, we say

$$\lim_{n \to \infty} x_n = x$$

or

$$x_n \to x$$

if, for every $\epsilon > 0$, there exists an N such that $|x_n - x| < \epsilon$ for all $n > N$.

Random Variables: Sure Convergence

$$X_n(\omega) \to X(\omega) \quad \forall \omega \in \Omega$$

Random Variables: Almost Sure Convergence

$X_n \stackrel{a.s.}{\to} X$ if and only if $P(\{\omega : X_n(\omega) \to X\}) = 1$, that is, X_n converges to X except possibly on a zero-measure set.

Does sure convergence imply almost sure convergence?

Random Variables: Convergence in Probability

$X_n \rightarrow_p X$ if $P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \rightarrow 0$ for any fixed $\epsilon > 0$.

Does $X_n \overset{a.s}{\rightarrow} X$ imply $X_n \rightarrow_p X$?

Random Variables: Convergence in Probability

Does $X_n \to_p X$ imply $X_n \overset{a.s}{\to} X$?

No.

PROOF. Consider the probability space $\Omega = (0, 1)$, with Borel σ-field and the Uniform probability measure U of Example $\boxed{1.1.11}$. Suffices to construct an example of $X_n \to_p 0$ such that fixing each $\omega \in (0, 1)$, we have that $X_n(\omega) = 1$ for infinitely many values of n. For example, this is the case when $X_n(\omega) = \mathbf{1}_{[t_n, t_n+s_n]}(\omega)$ with $s_n \downarrow 0$ as $n \to \infty$ slowly enough and $t_n \in [0, 1-s_n]$ are such that any $\omega \in [0, 1]$ is in infinitely many intervals $[t_n, t_n + s_n]$. The latter property applies if $t_n = (i-1)/k$ and $s_n = 1/k$ when $n = k(k-1)/2 + i$, $i = 1, 2, \ldots, k$ and $k = 1, 2, \ldots$ (plot the intervals $[t_n, t_n + s_n]$ to convince yourself). ∎

But there exists a subsequence n_k such that $X_{n_k} \overset{a.s.}{\to} X$.

Random Variables: Weak Convergence

$X_n \stackrel{D}{\to} X$ if $F_{X_n}(\alpha) \to F_X(\alpha)$ for each fixed α that is a continuity point of F_X.

"Weak convergence", "convergence in distribution", and "convergence in law" all mean the same thing.

Summary

Convergence:

Stronger

- Sure ("pointwise") $X_n \to X \iff X_n(\omega) \to X(\omega) \quad \forall\, \omega \in \Omega$
- Almost Sure $X_n \overset{a.s}{\to} X \iff P(\{\omega : X_n(\omega) \to X\}) = 1$
- In Probability

$$X_n \to_p X \iff P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \to 0 \quad \forall \epsilon > 0$$

- Weak ("in distribution"/"in law")

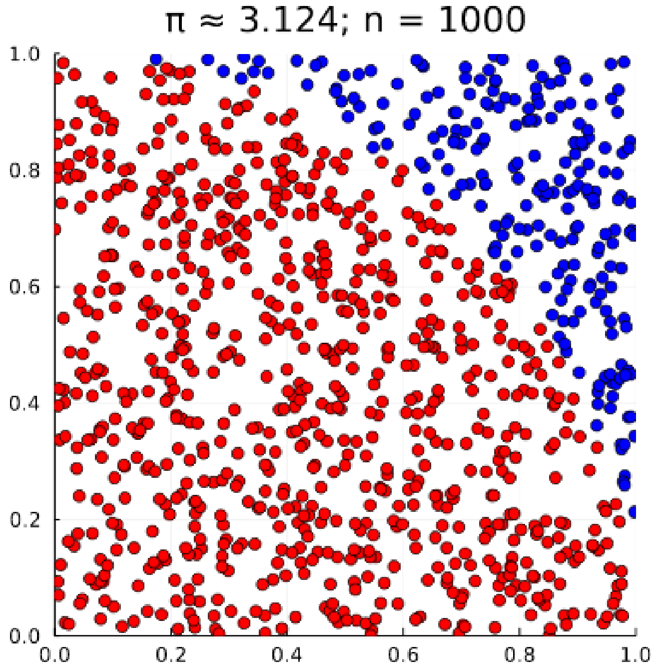
$X_n \overset{D}{\to} X \iff F_{X_n}(\alpha) \to F_X(\alpha)$ for each continuity point.

Implies

Break: Convergence of MC integration

$$X_i \sim U([0,1]^2)$$

$$Q_N \equiv \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{X_{i,1}^2 + X_{i,2}^2 \leq 1}(X_i)$$



(Intuitively)

$$Q_N \to \frac{\pi}{4} \text{ (sure)?}$$

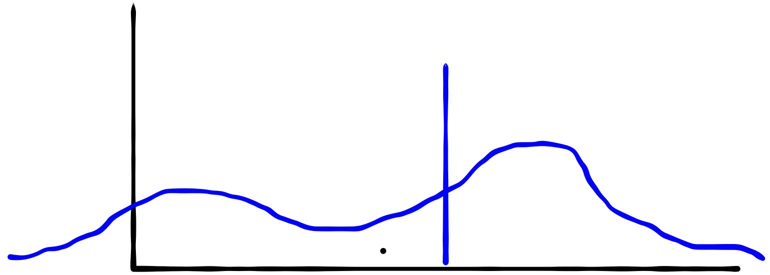
$$Q_N \overset{a.s.}{\to} \frac{\pi}{4}?$$

$$Q_N \to_p \frac{\pi}{4}?$$

$$Q_N \overset{D}{\to} \frac{\pi}{4}?$$

Concentration Inequalities

Intuition: If an r.v. has a finite variance, the probability that a random variable takes a value far from its mean should be small



Concentration inequalities take the form

$$P(X \geq t) \leq \phi(t)$$

where ϕ goes to zero (quickly) as $t \to \infty$

Concentration Inequalities

Markov's Inequality:

If $X \geq 0$, then

$$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall\, t \geq 0$$

Concentration Inequalities

Chebyshev's Inequality:

Let X be *any* real-valued random variable with $\mathrm{Var}(X) < \infty$. Then

$$P(|X - E[X]| \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}.$$

Very general, but still loose

Concentration Inequalities

Chebyshev's Inequality

k	Min. % within *k* standard deviations of mean	Max. % beyond *k* standard deviations from mean
1	0%	100%
$\sqrt{2}$	50%	50%
1.5	55.56%	44.44%
2	75%	25%
$2\sqrt{2}$	87.5%	12.5%
3	88.8889%	11.1111%
4	93.75%	6.25%
5	96%	4%
6	97.2222%	2.7778%
7	97.9592%	2.0408%
8	98.4375%	1.5625%
9	98.7654%	1.2346%
10	99%	1%

68% For Normal Distribution

95%

99%

Concentration Inequalities

Moment generating function: $M_X(t) \equiv E[e^{tX}]$

Chernoff Bound: If the moment-generating function M_X exists, then

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall\, t > 0$$

and

$$P(X \leq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall\, t < 0$$

Tighter than Markov and Chebyshev

Bonus Slide: Moment Generating Functions

To get the nth moment from $M_X(t)$, differentiate it n times and set $t = 0$.

Because:

$$
\begin{aligned}
M_X(t) = \mathrm{E}(e^{t\,X}) &= 1 + t\,\mathrm{E}(X) + \frac{t^2\,\mathrm{E}(X^2)}{2!} + \frac{t^3\,\mathrm{E}(X^3)}{3!} + \cdots + \frac{t^n\,\mathrm{E}(X^n)}{n!} + \cdots \\
&= 1 + tm_1 + \frac{t^2 m_2}{2!} + \frac{t^3 m_3}{3!} + \cdots + \frac{t^n m_n}{n!} + \cdots,
\end{aligned}
$$

Concentration Inequalities

Name	Requirements	Bound
Markov	$X \geq 0,\ \mathrm{E}[X]$ exists	$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall t \geq 0$
Chebyshev	$\mathrm{Var}(X) < \infty$	$P(\lvert X - E[X]\rvert \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}$
Chernoff	M_X exists	$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall t > 0$ $P(X \leq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall t < 0$

More restrictive

Tighter

Exercise

Name	Requirements	Bound
Markov	$X \geq 0$, $\mathrm{E}[X]$ exists	$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall\, t \geq 0$
Chebyshev	$\mathrm{Var}(X) < \infty$	$P(\lvert X - E[X] \rvert \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}$
Chernoff	M_X exists	$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall\, t > 0$ $P(X \leq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall\, t < 0$

Let Y be a r.v. that takes values in $[-1, 1]$ with mean -0.5. Give an upper bound on the probability that $Y \geq 0.5$.

(Weak) Law of large numbers

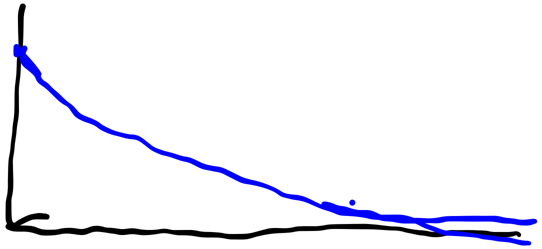
Let X_i be independent identically distributed r.v.s with mean μ and variance σ^2. If $Q_N \equiv \frac{1}{N}\sum_{i=1}^{N} X_i$, then $Q_N \to_p \mu$.

<u>Proof:</u>

Law of Large Numbers

Two somewhat astounding takeaways:

1. Standard deviation decays at $\frac{1}{\sqrt{N}}$ *regardless of dimension*.



2. You can estimate the "standard error" with

$$SE = \frac{s}{\sqrt{N}}$$

where s is the sample standard deviation.